Poisson Regression

Estimation, Interpretations and Diagnostics

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Last week: Logistic Regression

- Logistic regression: definition for binary responses.
- Estimation (MLE) and Interpretation (about β_1) for different type of covariates
- Inference: confidence intervals and tests
- Fitted values and residuals
- Diagnostics: overdispersion

similar to linear regression models except that the covariate effects are on log(odds) of the response, rather than the response itself.

$$\rightarrow E(y) = P(y=1) = \frac{e^{\beta_0+\beta_1 x}}{1+e^{\beta_0+\beta_1 x}}$$

$$Var(y) = E(y)(1-E(y))$$ under Bernoulli dist. assumption.

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This week: Poisson Regression

Poisson regression is a regression analysis designed for modelling **count data**.

The response variable represents the number of times an event occurs in a given timeframe or even in a space such as a geographic unit. For example,

- the number of customer arrivals
- disease occurrences
- traffic accidents

Poisson dist. $P(Y=k) = \frac{e^{-\lambda} \times \lambda^k}{k!}$, $k = 0, 1, 2, 3, \ldots$

$E(Y) = \lambda$

$Var(Y) = \lambda$

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The response

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Poisson is a common distribution to model a random variable that takes discrete non-negative integer values (i.e., 0, 1, 2,…)

$$Y_i|\mathbf{X}_i \sim \text{Poisson}(\lambda_i),$$

where λ_i is the conditional mean.

A particularity of the Poisson distribution is that its mean equals its variance:

$$\lambda_i = E[Y_i|\mathbf{X}_i] = Var[Y_i|\mathbf{X}_i]$$

Thus, any factor that affects the mean will also affect the variance.

> This fact could be a potential drawback for using a Poisson regression model.

Poisson regression can also be used to model *rates* of events but we won't cover that case in this course.

linear model
$Var(Y_i|X_i) = \sigma^2$
(indep. of β's)

Poisson model
$E(Y_i|X_i) = \beta_0 + \beta_1 X_i$
$Var(Y_i|X_i) = \beta_0 + \beta_1 X_i$

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Dataset

In this lecture, we'll use the `Bikeshare` dataset from the `ISL` package discussed in the ISL book

counts

The dataset contains information on the hourly users of a bike sharing program in Washington DC, for different months, hours, weather conditions, and membership types (see `?Bikeshare`)

```
The Bikeshare dataset has information about 8645 users
```

season	mnth	day	hr	holiday	weekday	workingday	weathersit	temp	ate
3	Aug	220	6	0	1	1	clear	0.70	0.6
4	Oct	293	21	0	4	1	clear	0.44	0.4

A range problem

Since **counts are nonnegative integers**, if we use linear regression, we might have a range problem by predicting negative counts.

Instead, we can use a *function* of the linear component with a positive range, for example, an exponential function.

> this is not the only function but it's commonly used due to its simplicity and interpretation

Poisson Regression

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The Poisson regression is defined as:

Y_i: counts

$$E[Y_i|\mathbf{X}_i] = \lambda_i = e^{\beta_0+\beta_1 X_{1,i}+\ldots+\beta_p X_{p,i}} \qquad \text{[mean]}$$

or equivalently,

→ linear predictor L_i

$$\log(\lambda_i) = \beta_0 + \beta_1 X_{1,i} + \ldots + \beta_p X_{\cancel{1,p}\ p,i} \qquad \text{[log-mean]}$$

not a r.v., so no e

> the log function links the conditional expectation with the linear component in a natural way in the GLM family so it's called the *canonical link*.

$E(Y_i)$

We model the log of the mean counts with a linear component!! → L_i

> then all that we learned for MLR can be used for this function of the conditional expectation!!

Note that there is no error term! we are modeling a function of the conditional expectation

$g(E(Y_i|X_i)) = L_i$
where $g(y) = \log(y)$
is a link function

(in a logistic model,
$g(y) = \log \frac{y}{1-y}$
in a linear model
$g(y) = y$))

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Poisson: a generalized linear model

The Poisson regression also belongs to the generalized linear models (GLMs) family.

In R, we can use the `glm()` function again but setting `family = poisson` to calculate the *maximum likelihood estimator* of the coefficients.

`glm()` models, by default, the log of the mean counts, which is the "canonical link" function but other functions can be used

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Fitting a Poisson Regression

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The interpretation of the coefficients will be very simular to those of the logistic regression case., except that the covariate effects are on log-scale insted of log(odds) scale.

So this time let's fit a model with various types of covariates at once.

From all the covariates available, let's consider

- X_1 `workingday`: an indicator variable that equals 1 if it is neither a weekend nor a holiday
- X_2 `temp`: the normalized temperature, in Celsius
- X_3 `windspeed`: the normalized windspeed
- X_4 `season`: a qualitative variable that takes four values for each season

categorical variable with 4 levels.

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Factors vs numeric

Note that R will create dummy variables of factors, not of numerical variables.

> sometimes numerical values are used for qualitative variables, e.g., season.

```
# A tibble: 2 × 5
  season workingday  temp windspeed bikers
   <dbl>      <dbl> <dbl>     <dbl>  <dbl>
1      1          0  0.26     0.134     34
2      1          1  0.28     0.104     96
```

You will need to adjust that before fitting the model

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Let's fit a Poisson model to the data as is. → include all predictors

```
summary(glm(formula = bikers ~ ., data = bikeshare,
            family = 'poisson'))
```

```
Call:
glm(formula = bikers ~ ., family = "poisson", data = bikeshare)

Coefficients:
             Estimate Std. Error z value Pr(>|z|)
(Intercept)  3.411649   0.004158  820.39   <2e-16 ***
season       0.123221   0.000969  127.16   <2e-16 ***
workingday  -0.028971   0.001940  -14.93   <2e-16 ***
temp         2.050466   0.004919  416.82   <2e-16 ***
windspeed    0.822224   0.007361  111.70   <2e-16 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

(Dispersion parameter for poisson family taken to be 1)

    Null deviance: 1052921  on 8644  degrees of freedom
Residual deviance:  805356  on 8640  degrees of freedom
AIC: 858404
```

← X should have 3 estimates for "season"

← ? If this is not true, the above results are wrong!

> Note that `workingday` and `season` were fit as numerical variables, and both should be treated as categorical variables!! ✓

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R can't distinguish continuous from categorical variables if the variable is numeric.

Factors are required for qualitative variables!

```
bikeshare <- bikeshare %>%
             mutate(season = as.factor(season),
                    workingday = as.factor(workingday))
```

```
Call:
glm(formula = bikers ~ ., family = "poisson", data = bikeshare)

Coefficients:
              Estimate Std. Error z value Pr(>|z|)
(Intercept)   3.345358   0.003981  840.38   <2e-16 ***
season2       0.062916   0.003741   16.82   <2e-16 ***
season3      -0.141267   0.004320  -32.70   <2e-16 ***
season4       0.379421   0.003357  113.01   <2e-16 ***
workingday1  -0.036254   0.001942  -18.67   <2e-16 ***
temp          2.688047   0.007435  361.54   <2e-16 ***
windspeed     0.741449   0.007377  100.50   <2e-16 ***
---
```

→ 3 estimates for "season"

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log-mean vs mean counts

As before, we can interpret the "raw" results corresponding to log-mean counts and are interpretated as for MLR

three dummy variables for "season"

$$\log(E[\text{bikers}|\mathbf{X}]) = \beta_0 + \beta_1 \text{S2} + \beta_2 \text{S3} + \beta_3 \text{S4} + \beta_4 \text{W1} + \beta_5 \text{temp} + \beta_6 \text{wind}$$

the reference category is hidden

$S_2 = \begin{cases} 1 & \text{season 2} \\ 0 & \text{otherwise} \end{cases}$

$S_3 = \begin{cases} 1 & \text{season 3} \\ 0 & \text{otherwise} \end{cases}$

for simplicity, S = season, W = workingday, and **wind** = windspeed, and $\mathbf{X}$ = all covariates

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Interpretation: log-mean

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term	estimate	std.error	statistic	p.value
(Intercept)	3.345	0.004	840.384	0
season2	0.063	0.004	16.818	0
season3	-0.141	0.004	-32.699	0
season4	0.379	0.003	113.008	0
workingday1	-0.036	0.002	-18.672	0
temp	2.688	0.007	361.541	0
windspeed	0.741	0.007	100.505	0

iClicker: true or false

should be log mean

1. Keeping other variables in the model constant, at any value, the mean bike usage decrease 0.036 in working days

   A: true B: false

2. Keeping other variables in the model constant, at any value, a unit increase in the normalized temperature is associated with an increase in log mean bike usage of 2.688.

   A: true B: false

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Interpretation: log-mean

The coefficients can be interpreted in terms of the log-means:

- **dummy variables** $\hat{\beta}_1 = 0.063$: the log-mean usage is season 2 is 0.063 larger than that in season 1 (reference), keeping all other variables in the model constant at any value

- **continuous variables** $\hat{\beta}_5 = 2.688$: an increase of 1 unit in `temp` is associated with an increase in the log-mean usage of 2.688, keeping all other variables in the model constant at any value

> same as for MLR but with response "log-mean counts"!

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Keeping constant …

$$\log \lambda_{\texttt{temp}} = \beta_0 + \beta_5 \overbrace{\mathrm{T}}^{\texttt{temp}} + \overbrace{\beta_1 \mathrm{S2} + \beta_2 \mathrm{S3} + \beta_3 \mathrm{S4} + \beta_4 \mathrm{W1} + \beta_6 \mathrm{wind}}^{\text{Constant}}$$

$$\log \lambda_{\texttt{temp + 1}} = \beta_0 + \beta_5 \overbrace{(\mathrm{T}+1)}^{\texttt{temp+1}} + \overbrace{\beta_1 \mathrm{S2} + \beta_2 \mathrm{S3} + \beta_3 \mathrm{S4} + \beta_4 \mathrm{W1} + \beta_6 \mathrm{wind}}^{\text{Constant}}$$

We take the difference between both equations as:

$$\begin{aligned} \log \lambda_{\texttt{temp + 1}} - \log \lambda_{\texttt{temp}} &= \beta_5 (\mathrm{T}+1) - \beta_5 \mathrm{T} \\ &= \beta_5 \end{aligned}$$

Thus, we interpret these coefficients as difference in log-mean counts

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Interpretation: mean counts

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But by logarithm property for a ratio:

$$\log \frac{\lambda_{\texttt{temp + 1}}}{\lambda_{\texttt{temp}}} = \log \lambda_{\texttt{temp + 1}} - \log \lambda_{\texttt{temp}}$$
$$= \beta_5$$

Poisson model:

$\log(E(y)) = \beta_0 + \beta_1 X_1 + \beta_2 X_2$

$E(y) = e^{\beta_0} \times e^{\beta_1 X_1} \times e^{\beta_2 X_2}$

Finally, we have to exponentiate the previous equation:

$$\frac{\lambda_{\texttt{temp + 1}}}{\lambda_{\texttt{temp}}} = e^{\beta_5}$$

This expression indicates that an increase of 1 unit in a continuous covariate is associated with a mean count change by a factor of e^{β_5} (multiplicative change).

$$\lambda_{\texttt{temp + 1}} = e^{\beta_5} \lambda_{\texttt{temp}}$$

> A more natural way of interpreting the coefficients of a Poisson regression

Using `tidy`: `exponentiate = TRUE`

```
#Coefficients for the mean counts
tidy(bikers_poisson, exponentiate = TRUE) %>%
  mutate_if(is.numeric, round, 3) %>%
  knitr::kable()
```

term	estimate	std.error	statistic	p.value
(Intercept)	28.371	0.004	840.384	0
season2	1.065	0.004	16.818	0
season3	0.868	0.004	-32.699	0
season4	1.461	0.003	113.008	0
workingday1	0.964	0.002	-18.672	0
temp	14.703	0.007	361.541	0
windspeed	2.099	0.007	100.505	0

e^{β_j}

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- **dummy variables** $e^{\hat{\beta}_1} = 1.065$: the mean usage is season 2 increases by a factor of 1.065 that in season 1 (reference), keeping all other variables in the model constant at any value
  - or by $(1.065 - 1) \times 100\% = 6.5\%$

- **continuous variables** $e^{\hat{\beta}_5} = 14.703$: an increase of 1 unit in `temp` is associated with an increase in mean usage of a factor of 14.703, keeping all other variables in the model constant at any value

iClicker:

dummy variables S3 $e^{\hat{\beta}_2} = -0.141 = 0.868$:

A: a change in season from season 1 to season 3 is associated with a change in mean bike usage by a factor of 0.868, keeping all other variables in the model constant at any value

B: only 86.8% as many people will use bikes in season 3 relative to season season 1, keeping all other variables in the model constant at any value

C: a change in season from season 1 to season 3 is associated

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Again, 4 Poisson regressions in 1!! For the **additive** model:

$$\log \lambda = \beta_0 + \beta_1 \mathrm{S2} + \beta_2 \mathrm{S3} + \beta_3 \mathrm{S4} + \beta_4 \mathrm{W1} + \beta_5 \texttt{temp} + \beta_6 \mathrm{wind}$$

reference season

$$\log \lambda = \begin{cases} \beta_0 + \beta_4 \mathrm{W1} + \beta_5 \texttt{temp} + \beta_6 \mathrm{wind} & \text{if } \mathrm{S2} = \mathrm{S3} = \mathrm{S4} = 0 \\ (\beta_0 + \beta_1) + \beta_4 \mathrm{W1} + \beta_5 \texttt{temp} + \beta_6 \mathrm{wind} & \text{if } \mathrm{S2} = 1,\ \mathrm{S3} = \mathrm{S4} = 0 \\ (\beta_0 + \beta_2) + \beta_4 \mathrm{W1} + \beta_5 \texttt{temp} + \beta_6 \mathrm{wind} & \text{if } \mathrm{S3} = 1,\ \mathrm{S2} = \mathrm{S4} = 0 \\ (\beta_0 + \beta_3) + \beta_4 \mathrm{W1} + \beta_5 \texttt{temp} + \beta_6 \mathrm{wind} & \text{if } \mathrm{S4} = 1,\ \mathrm{S2} = \mathrm{S3} = 0 \end{cases}$$

dummy variables

> (optional but useful) try to write these in exponential form

Model with interaction

For simplicity, let's look at a model with only `workingday` and `temp` but interacting

> we are thinking that the "effect" of temperature will be different for leisure days compared to working days.

$$\log \lambda = \gamma_0 + \gamma_1 \mathrm{W1} + \gamma_2 \texttt{temp} + \gamma_3 \mathrm{W1} \times \texttt{temp}$$

Again, 2 Poisson regressions in 1 with **different** slopes!

Again, 2 Poisson regressions in 1 with **different** slopes!

$$\log \lambda = \gamma_0 + \gamma_1 \mathrm{W1} + \gamma_2 \texttt{temp} + \gamma_3 \mathrm{W1} \times \texttt{temp}$$

For the **log-mean counts**:

$$\log \lambda = \begin{cases} \gamma_0 + \gamma_2 \texttt{temp} & \text{if W1} = 0 \\ (\gamma_0 + \gamma_1) + (\gamma_2 + \gamma_3)\texttt{temp} & \text{if W1} = 1 \end{cases}$$

> different *slope* (γ_2 vs $(\gamma_2 + \gamma_3)$) , different *intercepts* (γ_0 vs $(\gamma_0 + \gamma_1)$)!

For the **mean counts**:

$$\lambda = \begin{cases} e^{\gamma_0} e^{\gamma_2 \texttt{temp}} & \text{if W1} = 0 \\ e^{(\gamma_0+\gamma_1)} e^{(\gamma_2+\gamma_3)\texttt{temp}} & \text{if W1} = 1 \end{cases}$$

in R

include interaction

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```
bikers_poisson_int <- glm(bikers~workingday*temp,
                          data = bikeshare,
                          family = poisson)

tidy(bikers_poisson_int, exponentiate = TRUE) %>%
  mutate_if(is.numeric, round, 3) %>%
  knitr::kable()
```

term	estimate	std.error	statistic	p.value
(Intercept)	34.343	0.005	687.857	0
workingday1	1.481	0.006	63.450	0
temp	14.723	0.008	318.215	0
workingday1:temp	0.483	0.010	-71.201	0

significant interaction

Interpretation (for mean counts)

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In a **model with interactions**, the expected change in one variable **depends on** the value or levels the other variable

We **CAN'T** say "keeping the `temp` value constant" or "for working days or leisure days"

- *for leisure days*, a 1 unit increase in temperature is associated with an increase of mean bike usage **by a factor of** 14.723.

Interaction term

To get the "effect" of temperature for working days (i.e., `workingday` = 1), you need to *multiply* (instead of summing) the estimated coefficient of `temp` by that of the interaction term:

From the `tidy` table, with `exponentiate = TRUE`.

$$e^{\hat{\gamma}_2} = 14.723, \;\; e^{\hat{\gamma}_3} = 0.483$$

We need to multiply (instead of summing):

$$e^{\hat{\gamma}_2+\hat{\gamma}_3} = e^{\hat{\beta}_2} \;\; e^{\hat{\beta}_3} = 14.723 \times 0.483 = 7.11$$

In working days, a unit increase in temperature is associated with an increase of mean usage **by a factor of** 7.11.

Inference

Inference for Poisson regression is the same as that of Logistic regression. They are based on theorectical results from the CLT.

> if the sample size is large, the *Normal* distribution is used to obtain boudaries of confidence intervals and p-values

CLT

$$\frac{\hat{\beta}_j}{\widehat{\mathrm{SE}}(\hat{\beta}_j)} \sim N(0, 1)$$

> This is called the **Wald's statistic**

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The confidence intervals can be calculated for both the *log-mean* and the *mean* counts.

> options(digits = 2)

But again, R will do all this for you!

reject Ho

```
tidy(bikers_poisson,
    conf.int = TRUE, conf.level = 0.9) %>%
  mutate_if(is.numeric, round, 3) %>%
  knitr::kable()
```

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term	estimate	std.error	statistic	p.value	conf.low	conf.high
(Intercept)	3.345	0.004	840.384	0	3.339	3.352
season2	0.063	0.004	16.818	0	0.057	0.069
season3	-0.141	0.004	-32.699	0	-0.148	-0.134
season4	0.379	0.003	113.008	0	0.374	0.385
workingday1	-0.036	0.002	-18.672	0	-0.039	-0.033
temp	2.688	0.007	361.541	0	2.676	2.700
windspeed	0.741	0.007	100.505	0	0.729	0.754

all exclude 0

wrong

Residuals

The Poisson regression has the same problems as the Logistic Regression:

- responses are discrete (at least more than 2 values now)
- the variance is not constant

don't use this one

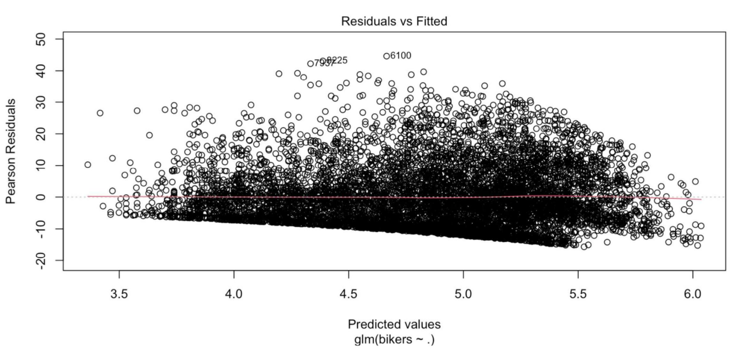
use this one

$$\text{resid}_{\text{raw}} = y_i - \hat{\lambda}_i, \qquad \text{resid}_{\text{pearson}} = \frac{y_i - \hat{\lambda}_i}{\sqrt{\hat{\lambda}_i}}$$

SE

Residual plot

```
plot(bikers_poisson,1)
```



Overdispersion

The variance of a Poisson random variable equals the mean, which in practice may not be observed.

> Poisson regression usually exhibits overdispersion.

An easy way to check is to fit a model using the quasilikelihood method and check if the dispersion parameter is very different from 1.

Poisson models
$$\begin{cases} E(y) = \beta_0 + \beta_1 x \\ Var(y) = E(y) \end{cases}$$
here $\gamma = 1$

$E(y) = \beta_0 + \beta_1 x$

$Var(y) = \gamma * E(y)$

called dispersion parameter

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in R

fit1

```
bikers_quasipoisson <- glm(bikers ~ .,
                           data = bikeshare,
                           family = quasipoisson)

summary(bikers_quasipoisson)
```

drop1(fit1)

```
Call:
glm(formula = bikers ~ ., family = quasipoisson, data = bikeshare)

Coefficients:
            Estimate Std. Error t value Pr(>|t|)
(Intercept)  3.34536    0.03789  88.286  < 2e-16 ***
season2      0.06292    0.03561   1.767 0.077302 .
season3     -0.14127    0.04112  -3.435 0.000595 ***
season4      0.37942    0.03196  11.872  < 2e-16 ***
workingday1 -0.03625    0.01848  -1.962 0.049849 *
temp         2.68805    0.07077  37.981  < 2e-16 ***
windspeed    0.74145    0.07022  10.558  < 2e-16 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

(Dispersion parameter for quasipoisson family taken to be 90.60984)

    Null deviance: 1052921  on 8644  degrees of freedom
Residual deviance:  783300  on 8638  degrees of freedom
AIC: NA

Number of Fisher Scoring iterations: 5
```

more reliable than previous results.

very far from 1, so there is a serious over-dispersion problem, i.e. Poisson dist does not hold.

$\gamma = 90.609$

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Summary

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- A count response can be modelled using a Poisson Regression
- The (conditional) expectation of a count response is the (conditional) *mean counts*.
- The (conditional) expectation of a count response equals the (conditional) variance
- A LR can not be used to model the conditional expectation of a count response since its range extends beyond positive values
- Instead, one can model a function of the conditional mean counts. A common choice in Poisson regression is to use the *log* function

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Summary (cont.)

The interpretation of the coefficients depends on the type of variables and the form of the model:

The raw coefficients (of the log-mean count model) are interpreted as:

- log-mean counts of a reference group
- difference of log-mean counts of a treatment vs a control group
- association between unit change in the input and changes in log-mean counts

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Summary (cont.)

The exponentiated coefficients (of the mean count model) are interpreted as:

- mean counts of a reference group
- ratio of mean counts of a treatment vs a control group
- multiplicative changes in mean counts per unit change in the input

Summary (cont.)

- The estimated Poisson model can be used to make inference using the Wald's satistic: confidence intervals and tests
- There are different type of fitted values (log-mean counts or counts?) and residuals
- Overdispersion is a potential problem in Poisson regression. Using *quasilikelihood* approach usual helps.